VELTRIX IT SOLUTION

WEEK: 02

NAME: ASIF NAWAZ

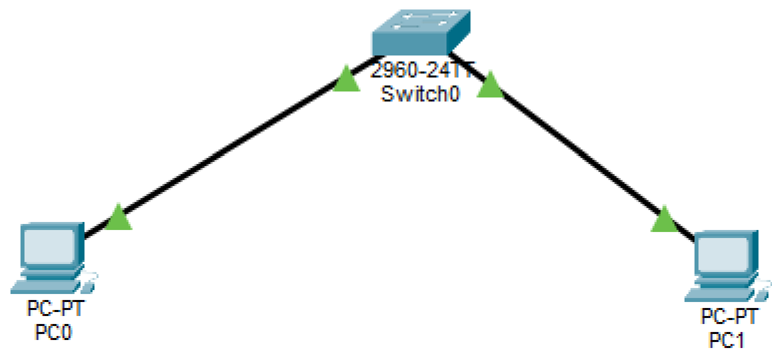
DOMAIN: NETWORK ADMINISTRATION

Day: 08

Practical:

- Ping between all devices in your Week 1 topology.

Topology



Now add IP addresses, PC1 192.168.10.1 and PC2 192.168.10.2. Now ping PC1 to PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Now Ping PC2 to PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Introduce a fault deliberately, then troubleshoot and fix it

To introduce a fault, we will change the IP address of PC2 and will ping it with PC1. The new IP of PC2 is 192.168.20.1.

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The ping is not done because there are two different IP addresses used in a single network.

Theory:

i. ICMP protocol and how ping works

ICMP:

- Internet Control Message Protocol
- It is a network-layer protocol used by devices like routers and computers to send error and messages and operational information.

Example:

- Checking if a device is reachable
- Reporting network errors
- Measuring delay

PING COMMAND:

The ping command uses ICMP to test connectivity between two devices.

1. Ping sends a request

Your computer sends an ICMP Echo Request packet to a target

2. Target Responds

If the target is reachable, it replies with an ICMP Echo Reply packet.

3. Time is measured

Your system calculates how long it took for the reply to come back. This is called RTT (round-trip time)

4. Result

You see:

- Time
- Packet loss
- Number of replies

Flow Chart:

Your Computer → Echo Request → Target Device

Target Device → Echo Reply → Your Computer

ii. What are the basic troubleshooting methods?

Basic Troubleshooting Methodology:

1. Identify the problem

- Start by understanding what's actually wrong.
- Ask questions: What happened? When did it start?
- Check error messages
- Try to reproduce the issue
- Identify scope (one user or many?)

2. Establish a Theory of Probable Cause

- Think of possible reasons behind the issue.
- Start with the simplest possibilities (cables unplugged, Wi-Fi off)
- Consider recent changes (updates, installs, config changes)

3. Test the Theory

- Now verify if your assumption is correct.
- Try a fix based on your theory
- If it works → great
- If not → go back and form a new theory

4. Establish a Plan of Action & Implement the Solution

- Once you know the cause:
- Plan the fix properly

- Consider risks (especially in networks/servers)
- Apply the solution

5. **Verify Full System Functionality**

- Don’t stop after “it works on my side.”
- Test everything affected
- Ensure no new issues were created
- Confirm with the user

Simple example:

Problem: Internet not working

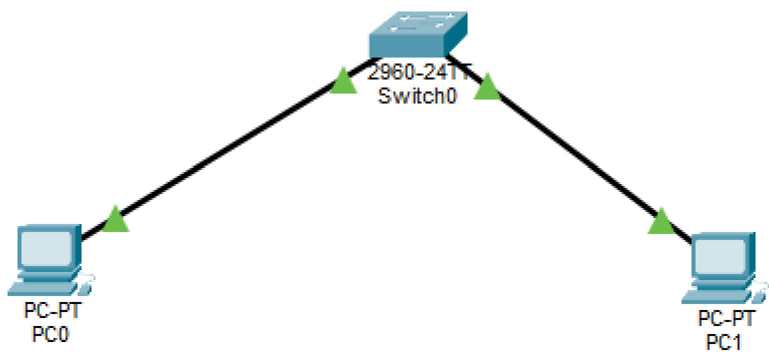
1. Identify → No websites loading
2. Theory → Wi-Fi disconnected
3. Test → Check Wi-Fi → It’s off
4. Fix → Turn Wi-Fi on
5. Verify → Internet works
6. Document → Done

Day: 09

Practical:

- Draw a VLAN segmentation diagram for an office. Create 2 VLANs in Packet Tracer and assign ports to each

Topology:



Assign Name to VLANs

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#
Switch(config-vlan)#
```

Assign access to the ports of the two PC's.

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switch
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#
```

The IP addresses used for both the PC's are from same network and still the ping not done. The reason is that both the PC's are from two different VLANs.

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Theory:

i. Why VLANs are used (segmentation, security, performance).

VLANs:

VLANs (Virtual LANs) are used to logically divide a single physical network into multiple smaller networks**.**

1. Segmentation:

- Used for Network Organization
- VLANs let you group devices based on function not location.

Example:

- VLAN 10 → HR
- VLAN 20 → IT
- VLAN 30 → Accounts

Even if all PCs are on one switch, each department behaves like it's on its own network.

2. **Security:**

Devices in one VLAN cannot directly communicate with another VLAN unless routing is configured.

Example:

- HR data stays isolated from IT
- Guest users can be placed in a separate VLAN

Benefits:

- Limits unauthorized access
- Reduces risk of attacks spreading
- Better control with access rules (ACLs)

3. **Performance:**

Devices in one VLAN cannot directly communicate with another VLAN unless routing is configured.

Example:

- HR data stays isolated from IT
- Guest users can be placed in a separate VLAN

Result:

- Less network congestion
- Faster communication
- More efficient bandwidth usage

ii. Difference between access ports and trunk ports

1. **Access Port:**

An access port belongs to only one VLAN.

- Connects to end devices (PCs, printers)
- Carries untagged traffic
- Each port = one VLAN

Example:

- PC connected to VLAN 10

interface fa0/1

switchport mode access

switchport access vlan 10

2. Trunk Port:

A trunk port carries multiple VLANs.

- Connects switches to switches OR switch to router
- Uses VLAN tagging (802.1Q)
- Can carry many VLANs at once

Example:

interface fa0/24

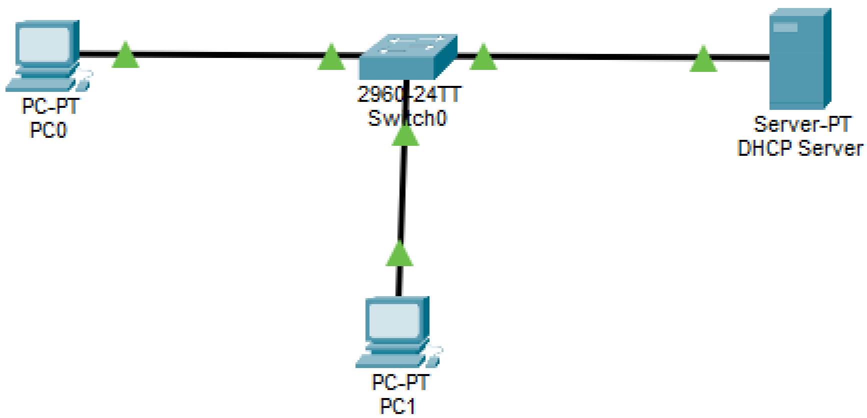
switchport mode trunk

Day: 10

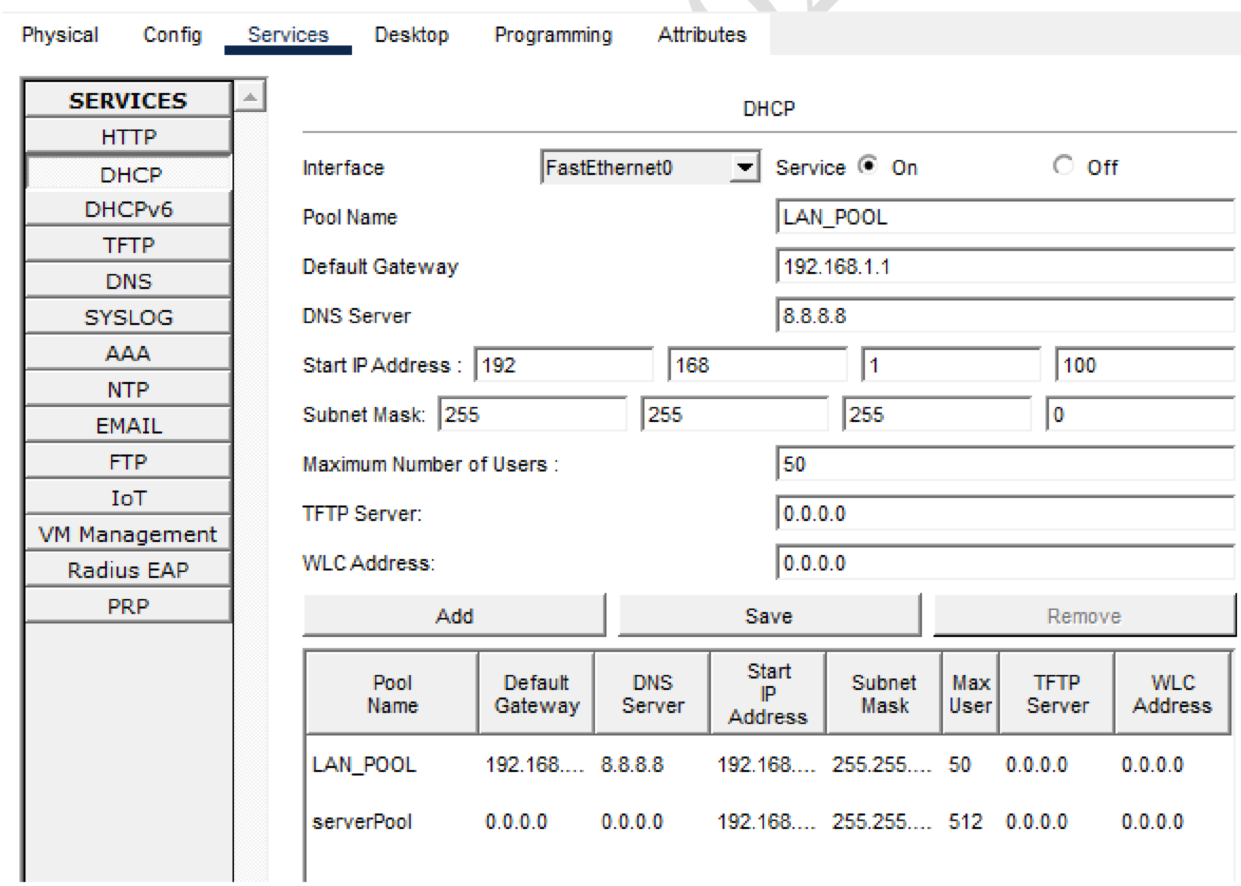
Practical:

- Configure a DHCP server in Packet Tracer. Verify auto IP assignment on client PCs.

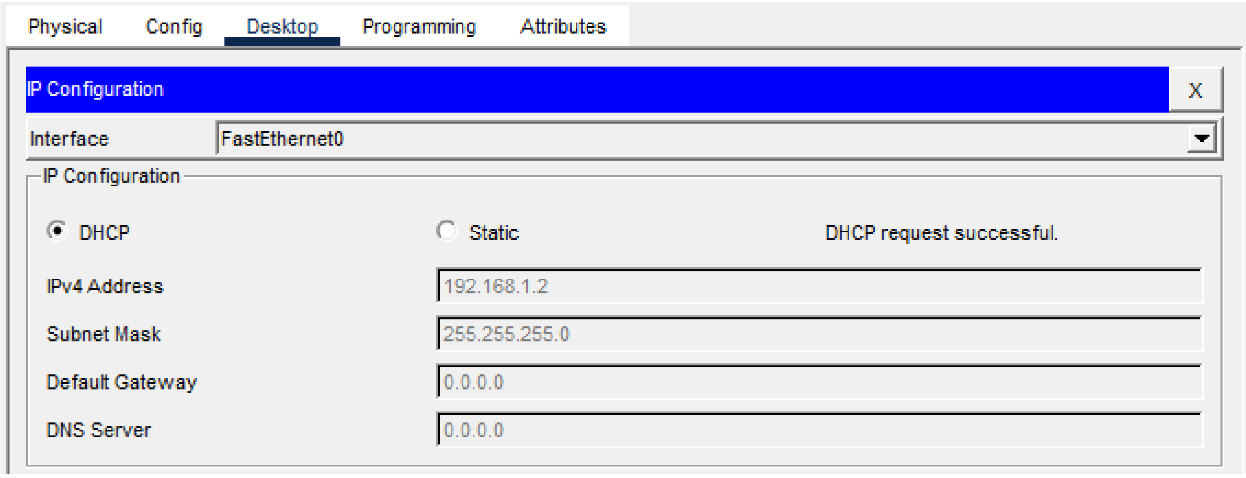
Topology:

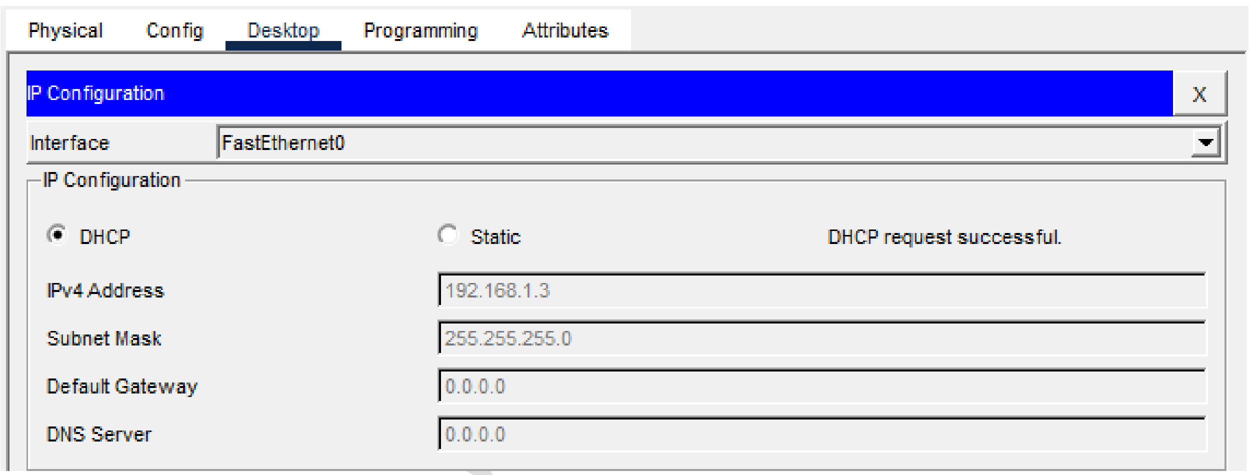


Open the DHCP server and assign IP address to it.

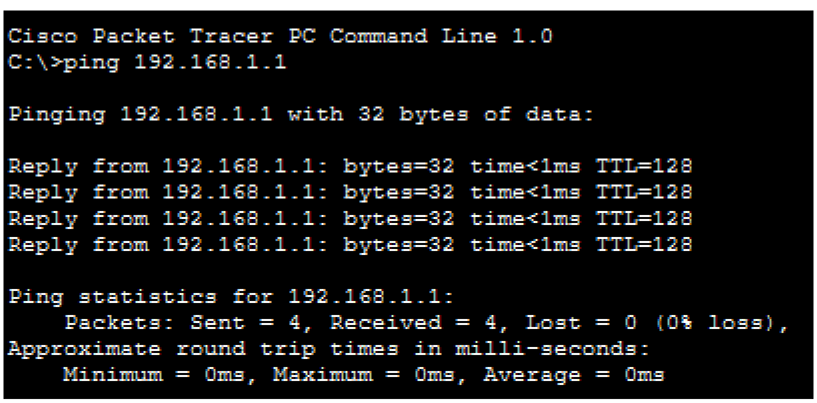


When the DHCP IP are added to the server and now you open the PCs IP Configuration and click DHCP, the PC will be automatically assign IP.





Ping from one PC to server or server to any PC, it will be pinged.



Theory:

- Static IP vs Dynamic IP: when to use each.

Static IP:

A static IP is configured manually on a device and doesn't change**.**

Uses:

- Servers (web, file, database)
- Network devices (routers, switches, printers)
- CCTV / NAS / critical systems
- Remote access or port forwarding needed

Dynamic IP:

A dynamic IP is assigned automatically by a DHCP server and can change over time**.**

Uses:

- End-user devices (PCs, mobiles, laptops)
- Large networks with many users
- Temporary or guest devices

- DHCP lease process (DORA: Discover, Offer, Request, Acknowledge)

When a device joins a network, it follows the DORA process to get an IP address.

DORA: Discover Offer Request Acknowledge

- **Discover:**

Client broadcast will ask, "Is there any DHCP Server available?"

- **Offer:**

DHCP server replies: "Yes, here's is an IP you can use"

- **Request:**

Client says: "I accept this IP address"

- **Acknowledge:**

Server confirms: "IP assigned successfully"

Simple Flow:

Client
↓
Discover
↓
DHCP Server
↓

Offer

↓

Client

↓

Request

↓

DHCP Server

↓

Acknowledge

↓

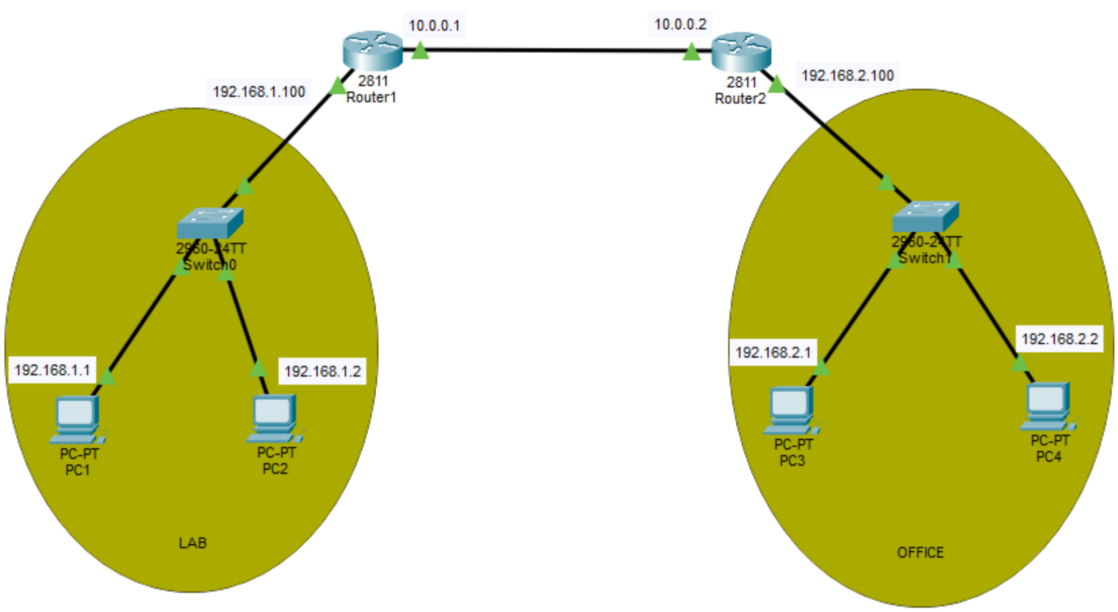
Client

Day: 11

Practical:

- Connect 2 different networks via a router in Packet Tracer. Configure static routes and verify inter-network ping.

Topology:



Router 1 Configuration:

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip add 192.168.1.100 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#int f0/1
Router(config-if)#
Router(config-if)#ip add 10.0.0.1 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

Router 2 Configuration:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip add 192.168.2.100 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip add 10.0.0.2 255.255.255.252 no sh
                                                   ^
% Invalid input detected at '^' marker.

Router(config-if)#ip add 10.0.0.2 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Ping PC1 to PC4:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping PC4 to PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=6ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

Theory:

- Static routing vs dynamic routing

Static Routing:

In static routing you manually configure routes on each router.

Uses:

- Small networks
- Simple topologies
- When you want full control

Advantages:

- Simple and predictable
- No extra CPU/bandwidth usage
- More secure /no route advertisements

Disadvantages:

- Not scalable
- Must update manually if network changes

Dynamic Routing:

Routers learn routes automatically using routing protocols like:

- RIP
- OSPF
- EIGRP

Uses:

- Large or growing networks
- Networks with frequent changes

Advantages:

- Automatically updates routes
- Scalable and flexible
- Find best path dynamically

Disadvantages:

- More complex
- Uses CPU and bandwidth
- Needs proper configuration

- How Routing Tables work

A routing table is like a router's map

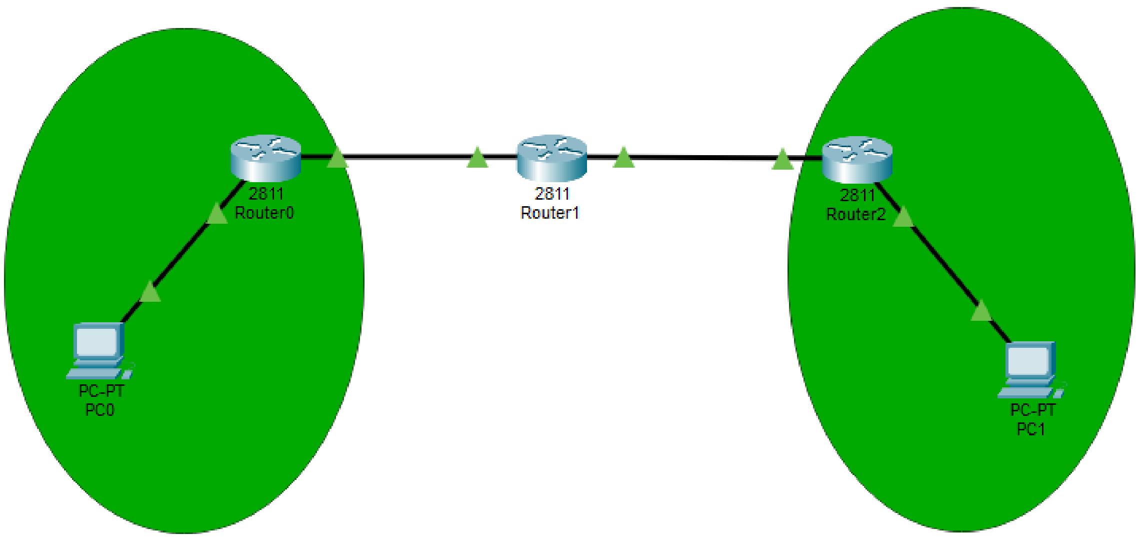
- It stores information about:
    - Destination networks
    - Next-hop IP addresses
    - Outgoing interfaces

Day: 12

Practical:

Configure RIP or OSPF on routers in Packet Tracer

Topology:



RIP Interfacing:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto
Router(config-router)#no auto-summary
Router(config-router)#
Router(config-router)#
```

```
Router(config)#int f0/0
Router(config-if)#ip add 192.168.1.0 255.255.255.0
Bad mask /24 for address 192.168.1.0
Router(config-if)#ip add 192.168.1.10 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip add 10.0.0.1 255.0.0.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

Theory:

- RIP vs OSPF: differences and use cases.

Routing Information Protocol (RIP):

Working:

- Uses hop count as metric
- Maximum 15 hopes
- Sends full routing table periodically

Characteristics:

- Very simple
- Slow to adapt
- Less scalable

Uses:

- Small networks
- Lab environments
- Simple brank networks

Open Shortest Path First (OSPF):

Working:

- Uses cost (bandwidth-based metric)
- Builds a full topology map (link-state database)
- Uses Dijkstra shortest path algorithm

Characteristics:

- Fast coverage
- Highly scalable
- More complex than RIP

Uses:

- Large enterprise networks
- ISP networks
- Modern-real-world routing systems

- Concept of convergence in dynamic routing

Coverage:

Coverage is the time it takes all routers in a network to agree on the same network topology after a change.

Explanation:

When something changes in the network:

- Link goes down
- New router added
- Path changes

Router must:

- Detect the change
- Update routing tables
- Share updates with others
- Agree on new best paths

The time taken is convergence time.

Example:

A link between R1 and R2 fails:

RIP:

Routers slowly update every 30 seconds.

May take time to remove bad route

OSPF:

Immediately detects change

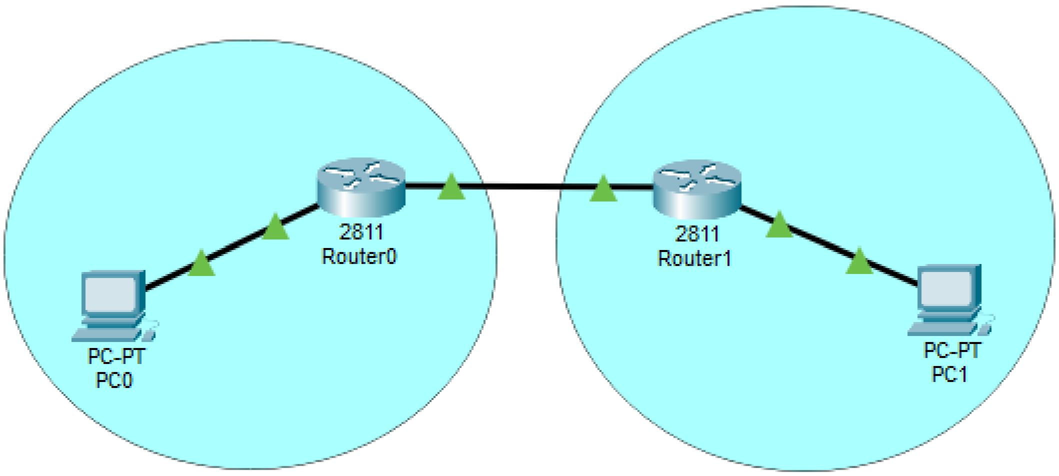
Recalculates new shortest path quickly

Day: 13

Practical:

Configure NAT on a border router in Packet Tracer. Test internet-facing connectivity through NAT.

Topology:



Configuring Router 1:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip route 0.0.0.0 0.0.0.0 1.1.1.2
Router(config)#int fa 0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip add 192.168.10.10 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
Router(config-if)#int fa 0/1
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#ip add 1.1.1.2 255.255.255.0
Router(config-if)#no sh
```

Configuring Router 2:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip add 192.168.20.10 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
Router(config-if)#int fa 0/1
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#ip add 1.1.1.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
```

The output is not responding because the NAT on Router one was configure and the ping from PC1 to PC2 not occurred.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Theory:

- How NAT translates private IPs to public IPs

NAT Translation:

- NAT (Network Address Translation) runs on a border router/firewall. It rewrites IP addresses so devices using private addresses can talk to the internet.
- Source IP is modified on the way out
- On the way back, NAT uses a translation table to restore the original private IP/port.

Simple Flow:

PC (192.168.1.10:5000) ──▶ Router ──▶ Internet

(Translated to 203.0.113.5:30001)

Internet reply ──▶ Router ──▶ PC

(203.0.113.5:30001 → 192.168.1.10:5000)

- Types of NAT: Static, Dynamic, PAT (Overload)

Types of NAT:

Static NAT:

- One private IP <-> One public IP
- Hosting a server that must be reachable from outside

Example:

192.168.1.10 <-> 203.0.113.10

Pros: predictable, always reachable

Cons: needs a dedicated public IP per device

Dynamic NAT:

- Private IPs are mapped to a pool of public IPs
- You have a small pool of public IPs and many internal hosts.

Example:

Pool: 203.0.113.10 – 203.0.113.20

192.168.1.10 → 203.0.113.10

192.168.1.11 → 203.0.113.11

Pros: automatic allocation

Cons: limited by pool size

PAT/NAT Overload:

- Many private IPs share a single public IP using different ports
- Typical home/ office internet access

Example:

192.168.1.10:5000 → 203.0.113.5:30001

192.168.1.11:5001 → 203.0.113.5:30002

Pros: very efficient, saves public IPs

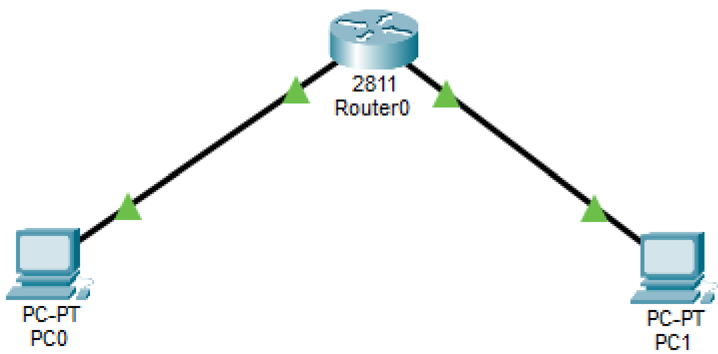
Cons: inbound access required port forwarding

Day: 14

Practical:

Apply Access Control List (ACL) rules on a router. Block specific traffic and verify the rule works.

Topology:



Configure ACL on the Router:

```
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny icmp host 192.168.10.1 host 192.168.20.1
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#int f0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

This will deny the request from first PC to the second PC. Now if we ping from PC1 to PC2, it will show destination host unreachable.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.10.10: Destination host unreachable.
Reply from 192.168.10.10: Destination host unreachable.
Reply from 192.168.10.10: Destination host unreachable.
Reply from 192.168.10.10: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

And ping from PC2 to PC1, it will show request time out. Because the traffic is deny on the f0/0 path connected from the PC1 to router.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Theory:

- What are the Network security fundamentals?

Network Security Fundamentals:

There are the basic principles and practices used to protect a network from unauthorized access, attacks, and data loss.

CIA Triad:

- Confidentiality: Only authorized users can access data
- Integrity: Data should not be altered
- Availability: Systems should be accessible when needed

Authentication and Authorization:

- Authentication: Verifies identity (login, password, biometrics)
- Authorization: Defines what a user is allowed to do

Access Control:

- Limit who can access network resources
- Use tools like:
  - Firewalls
  - ACLs (Access Control Lists)
  - VLANs

Encryption:

- Protects data by converting it into unreadable form
- Used in HTTPS, VPNs, etc.

Firewalls:

- Monitor and control incoming/outgoing traffic
- Block unauthorized access

- Standard ACL vs Extended ACL

Standard ACL:

Filters traffic based only on source IP address

Example:

access-list 1 permit 192.168.1.0 0.0.0.255

Characteristics:

- Simple
- Less control
- Usually placed near destination

Extended ACL:

Filter traffic bases on:

- Source IP
- Destination IP
- Protocol
- Port Numbers

Example:

access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80

Characteristics:

- More detailed control
- More secure
- Usually placed near source